

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Department of Mathematics

SI 427 (Probability I)

Tutorial Sheet-V

All random objects are in $(\Omega, \mathcal{F}, P)$ unless told otherwise.

1. Let X be a random variable with distribution function F . Show that

$$P\{X = x\} = F(x) - F(x-), \quad x \in \mathbb{R}.$$

2. An unbiased coin is tossed 3 times and the trials are independent. Let X denote the number of H 's. Find the distribution function of X .
3. Give an example of a random variable which is neither discrete nor continuous.
4. An X -coin is tossed twice independently, where X is a random variable taking values $\frac{1}{2}$ and $\frac{1}{3}$ with equal probability. Let Y denote the number of H 's. Find the distribution function of Y .
5. Let X be an exponential(1) random variable and denote the distribution function of X by F . Find the distribution of $F \circ X$.
6. Define $F : \mathbb{R} \rightarrow [0, 1]$ by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{x^2}{4} & \text{if } 0 \leq x \leq 2 \\ 1 & \text{if } x > 2. \end{cases}$$

Show that F is a distribution function. Does pdf exists?

7. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \frac{1}{2}e^{-|x|}, \quad x \in \mathbb{R}.$$

Show that f is a pdf of some random variable say X . Also find $P(1 \leq |X| \leq 4)$.

8. Let X be exponentially distributed with parameter λ . Find the pdf of cX , $c > 0$.

9. Let X be a random variable with distribution function F and $a \in \mathbb{R}$. Define $Y = \max\{X, a\}$. Find the distribution function of Y in terms of F .
10. Let X be a random variable with distribution function F . Find the distribution function of the following random variables in terms of F .
(i) X^2 (ii) $|X|$ (iii) e^X .
11. Let X be a continuous random variable with pdf f . Let X is symmetric¹ and X^2 is exponential with parameter λ . Find f .
12. Let X be a random variable with distribution function F and $Y = aX + b$. Find the distribution function of Y in terms of F .
13. Let X be Uniform $(-1, 1)$ random variable and $Y = X^2$. Find the pdf of Y .
14. Let X be uniform $(0, 1)$ random variable. Find $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g \circ X$ is uniform $(-1, 1)$.
15. Let X be a continuous random variable with distribution function F and $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is given by

$$\varphi(x) = \begin{cases} x - 1 & \text{if } x < -1 \\ 0 & \text{if } -1 \leq x \leq 1 \\ x + 1 & \text{if } x > 1. \end{cases}$$

Find the distribution of $\varphi \circ X$ in terms of F .

¹A random variable X is said to be symmetric if X and $-X$ have the same distribution.